

EFFECTS OF OFFICE LIGHTING ON MOOD AND COGNITIVE PERFORMANCE AND A GENDER EFFECT IN WORK-RELATED JUDGMENT

IGOR KNEZ is a lecturer in psychology at the Royal Institute of Technology, Gävle, Sweden. He has a Ph.D. from the University of Uppsala, and his research interests include environmental, cognitive, and emotional psychology.

INGELA ENMARKER is a Ph.D. student at the Uppsala University, Centre for Caring Sciences. Her research interests focus on cognitive gerontology.

ABSTRACT: The study presents an investigation of the effects of the recommended office lighting on subjects' mood and cognitive performance in the physical setting of an office. In addition, a gender effect in the performance appraisal task was examined, both as a between- and within-subject factor. The results showed no significant effect of the lighting on the performance of cognitive tasks. However, an interaction between gender and color temperature on mood showed that 3000K (more reddish) and 4000K (more bluish) office lighting may communicate different affective loadings or meanings to each gender. The cognitive workload induced by almost 2 hours of intellectual work diminished the subjects' positive mood and augmented a negative mood. Moreover, independently of their gender, the raters evaluated the *neutral* female significantly different from the *neutral* male ratee. Implications of these findings for the mood effects of indoor lighting and the gender effect in work-related judgment are discussed.

Compared to earlier epochs, present-day urban society generates a variety of activities within the walls of artificial milieus. One such activity is intellectual work that is expanding as a function of the growth of modern society. Given this, we could say that in some respects, the natural milieu of living of man today is an artificial one and that the cognitive operations of

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our intellectual life are daily at work in that kind of setting. How, then, does an artificial milieu affect our cognitive labor?

In the light of numerous findings showing effects (a) of physical variables on emotion, behavior, and performance (see, e.g., Bell, Fisher, Baum, & Greene, 1990; Cohen, Evans, Stokols, & Krantz, 1986; Russell & Snodgrass, 1987, for reviews) and (b) of emotion on cognition (see, e.g., Clark & Fiske, 1982; Isen, 1993; Kuiken, 1991, for reviews), Knez (1995a) sketched an experimental research on the effects of artificial light on mood and cognition. The main hypothesis was that moods induced in subjects by the recommended level of office lighting (Commission Internationale De L'Eclairage [CIE], 1986) may, in turn, influence the ongoing cognitive performance employed in that artificial setting.

Baron, Rea, and Daniels (1992) tested for a similar effect of indoor lighting on cognitive performance via positive mood. In line with earlier results (e.g., Boray, Gifford, & Rosenblood, 1989), they did not show direct effects of the artificial light on mood. Knez (1995a) argued that the reason for this may be twofold. First, the lighting exposure was not of sufficient duration to produce a mood response. Second, mood data were not collected before and after the exposure so as to enable calculation of change scores for positive and negative moods, which could serve as dependent variables in the statistical analyses.

Given this, Knez (1995a) designed two laboratory lighting-exposure experiments that varied gender and three parameters of lighting in a factorial between-subject design: (a) color temperature—the type of lamp, measured in Kelvin (K); (b) illuminance—the level of lighting, measured in lux; and (c) color rendering index (CRI)—an index with point 100 as the best color appearance produced by a lamp at each respective color temperature. The experiments lasted for approximately 2 hours, and moods were measured before and after the exposure.

The artificial light did change mood in subjects, and this either facilitated or impaired their cognitive processes. This was true for long-term memory and problem-solving performances but not for judgment (Knez, 1995a). For example, the lighting that induced the least negative mood in a gender facilitated this gender's performance in problem solving, long-term recall, and recognition tasks (Knez, 1995a, Experiment 1). In addition, the free recall

Löfberg, Terry Hartig, and the two anonymous reviewers for their constructive criticism. Address correspondence to Igor Knez, Royal Institute of Technology, Centre for Built Environment, Laboratory of Applied Psychology, Box 88, 801 02 Gävle, Sweden; phone: +46-26-14-78-81, fax: +46-26-14-78-01, e-mail: Igor.Knez@bmg.kth.se.

results supported the mood-congruence hypothesis (see, e.g., Blaney, 1986, for a review); in accordance with their positive mood, subjects recalled more words from a positive than a negative word list.

As mentioned, the results in the judgment task did not show a mood-mediated effect of artificial light on cognition. The judgment was measured by the performance appraisal task (see Cascio, 1978, for a review). In this case, the task involved only neutral (balanced) information about an appraisee (Baron et al., 1992). This was employed to test the mood-congruence hypothesis in judgment: Does an appraiser judge a neutral appraisee (a fictitious employee) in accordance with his or her positive or negative mood, either overestimating when in a positive mood or underestimating when in a negative mood? Knez (1995a) showed that subjects were mostly perceived to be in a positive mood and that they did, indeed, overestimate the neutral ratee in accordance with their positive mood. Yet, this was true only for males. Given that the fictitious employee under judgmental scrutiny was a female secretary, Knez (1995a) tentatively interpreted this result as showing a halo effect (e.g., Cooper, 1981), that is, "a tendency to allow an overall impression of a person or one particular outstanding trait to influence the total rating of that person" (Reber, 1985). Consequently, in this case, "an overall impression . . . or one particular outstanding trait" was the gender of that ratee.

Given all this, the present objective is twofold, namely, to investigate (a) the mood-mediated effects of the recommended office lighting on subjects' intellectual performance, as indirectly shown in Knez (1995a), and (b) gender effects in a performance appraisal task.

One way to investigate the gender effect conclusively is to vary gender both as a between- and a within-subject independent variable in an experimental design (Cooper, 1981). Thus, in the present performance appraisal task, male and female raters (between-subject variable) judged skills (dependent variable) in both male and female ratees (within-subject variable).

As regards the physical stimuli, the present experiment varied the color temperature (type of lamp) and kept the illuminance and the CRI parameters constant. This was done to replicate Knez (1995a, Experiment 1), which showed that 3000K (more reddish) and 4000K (more bluish) lighting at high illuminance (1500 lux) and at high CRI (95) accounted for induction of the least negative mood in each gender, respectively, and so enhanced their performance in problem solving, long-term recall, and recognition tasks.

METHOD

PARTICIPANTS

Eighty participants were recruited from a local college of health sciences. Twenty males and 20 females were randomly assigned to each of the two luminous milieus. They ranged in age from 18 to 55 and were paid approximately US\$30 for participating.

EXPERIMENTAL SETTING

The study was performed in a chamber (3.9 m $\times$ 3.8 m $\times$ 2.5 m) in which the air temperature and humidity were controlled by a computerized climate system. The room was off-white and furnished as an office without any exaggerations in design. Two to four subjects were run at the same time. Six ceiling-mounted luminaries with four fluorescent tubes each were installed in the room. The color temperatures were applied by changing the tubes: (a) 3000K (Osram, 36W, L36/32, CRI = 95) and (b) 4000K (Osram, 36W, L36/22, CRI = 95). The lamps were on for 20 to 30 minutes before each session to stabilize the illuminance level (1500 lux). The color temperature, CRI, and illuminance levels were all in accordance with international recommendations (CIE, 1986), as far as they concern office settings and the tasks performed in that kind of milieu.

BASIC DESIGN AND INDEPENDENT VARIABLES

A factorial between-subject design was employed with two independent variables: 2 (color temperature, 3000K vs. 4000K) $\times$ 2 (genders).

DEPENDENT MEASURES

The subjects had to accomplish three memory tasks, a problem-solving task, a judgment task, and complete a mood and a room light questionnaire.

Long-term recall and recognition. The material to be remembered consisted of a dense, compact, 7-page text about an ancient culture (Carter, 1982). The text was handed out at the beginning of the experiment. At the end of the experiment, subjects were asked to accomplish a recall and a recognition task consisting of six general knowledge questions and 18 multiple choice questions, respectively (Hygge, 1993).

Free recall. Three word lists containing 16 words each and colored with neutral, positive, and negative hedonic tones were used as the to-be-remembered material. They were presented on a computer screen, randomly across subjects. After each word-list presentation, subjects were asked to write down all the words they recalled from the list but not in any particular order (Knez, 1995a).

Problem solving. The embedded figure task (e.g., Hartley, 1989) was used as a measure of problem-solving processes. This task comprised two types of figures. At the top of the answer sheet, there were five small solution figures (A, B, C, D, and E) and below them a large and complex target figure. The subjects' task was to find one of the five small figures within the complex one. The search for a problem solution (one of the five small figures) was ended by marking the corresponding capital letter below the target figure. All together, there were 32 target figures divided into two answer sheets.

Judgment. A performance appraisal task (Baron et al., 1992) was made up of neutral (balanced) information about a fictitious employee and 10 rating scales: qualifications for a current job, work-related skills, motivation, attitude to the job, work-related achievement, intelligence, concern with the work-related instructions, ability to get along with others, deserves credit, and an overall evaluation. Judgments were made on a 7-point scale.

To test the gender effect, subjects had to accomplish two performance appraisal tasks. They were identical, except one contained information about a male bank clerk and the other information about a female bank clerk. The information about these two bank clerks and the order of succession of the rating scales were superficially rewritten and reordered between the tasks (bank clerks). This was done to give the impression that the female and the male bank clerks, respectively, yielded two different rates. The bank clerk profession was used because in Sweden, both genders are equally represented in this profession and, generally speaking, Swedes have the same kind of attitudes toward male and female bank clerks. Furthermore, the administration of the two versions of this task was balanced across subjects; half of the subjects rated the male bank clerk first and the other half rated the female bank clerk first.

Mood. The Positive and Negative Affect Schedule (PANAS) (Watson, Clark, & Tellegen, 1988) is composed of 20 items that represent two orthogonal dimensions of mood (positive vs. negative). Ratings are made on a 5-point scale (from *little or not at all* to *very much*) with respect to the question: "How do you feel now?"

Room light evaluation. This questionnaire was evolved to investigate the subjects' perceived estimations of the room light (Knez, 1995a). It was composed of seven items, strictly related to the quality of the artificial light, namely, glaring, dim, soft, bright, warm, intense, and cool. The ratings were made on a 5-point scale (from *little, or not at all* to *very much*) related to the question: "How would you evaluate the room light?"

PROCEDURE

Two to four subjects were tested at each session. They were told that the study was about people's memory, problem-solving, and judgment processes. For each subtask, subjects received instructions concerning that particular task and then were given a time limit to accomplish that task. This was done to implement an even time exposure of the room lighting across all conditions and subjects. Altogether, the experiment lasted approximately 140 minutes. The tasks were given in the following order: (a) PANAS (Block 1), 5 minutes; (b) text reading (encoding), 35 minutes; (c) problem-solving task, 35 minutes; (d) free recall task, 5 minutes; (e) PANAS (Block 2), 5 minutes; (f) performance appraisal task, one (male or female bank clerk), 15 minutes; (g) long-term recall and recognition tasks (long-term retrieval, text questions), 20 minutes; (h) performance appraisal tasks, two (male or female bank clerk), 15 minutes; and (i) room light evaluation questionnaire, 5 minutes.

RESULTS

MOOD

Two analyses of variance were computed. The first one comprised the basic design independent variables, with the Mood-Block 2 minus Mood-Block 1 delta (change) scores as the dependent variable. The second analysis was computed on subjects' mood scores that entailed the following independent variables: 2 (genders) $\times$ 2 (color temperatures) $\times$ 2 (mood-blocks), with the last independent variable as a within-subject factor.

Positive mood. The delta score analysis showed a strong tendency to a significant two-way interaction between gender and color temperature, $F(1, 76) = 3.39, p = .06$. As can be seen in Figure 1, the best preservation of positive mood was generated by the 3000K lighting for males and by the 4000K lighting for females.

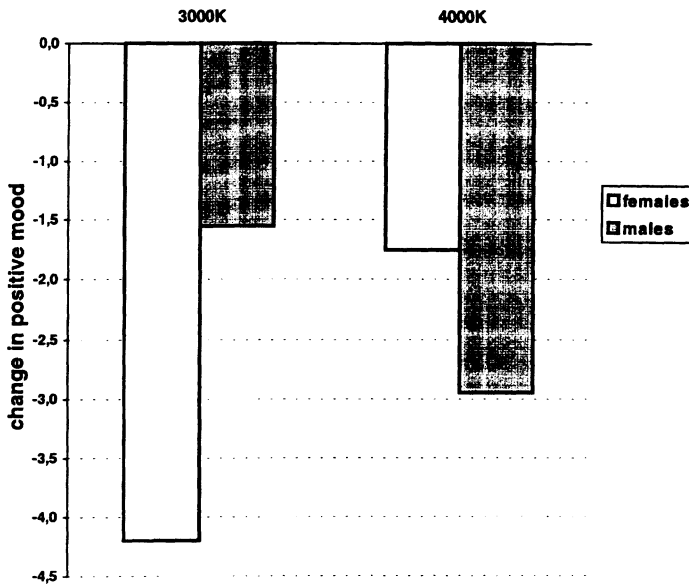


Figure 1: Mean Change in Positive Mood in Females and Males as a Function of 3000K (more reddish) and 4000K (more bluish) Lighting

The second analysis of variance showed a main effect of mood-block, $F(1, 76) = 24.99, p < .01$. The positive mood diminished in subjects over mood-blocks, albeit still high (from 30.2 to 27.6 on a scale ranging from 10.0 to 50.0). It must be noted that only the second analysis of variance permitted a main effect of mood-block.

Negative mood. The second analysis showed a main effect of mood-block, $F(1, 76) = 4.52, p < .05$ (negative mood increased over mood-blocks, from 11.9 to 12.7), and a two-way interaction between gender and color temperature, $F(1, 76) = 4.48, p < .05$. As can be seen in Figure 2, the 3000K lighting induced the least negative mood in males and the 4000K lighting accounted for the same effect in females.

MEMORY AND PROBLEM-SOLVING TASKS

Memory. No significant results were obtained in free recall, long-term recall, and recognition tasks.

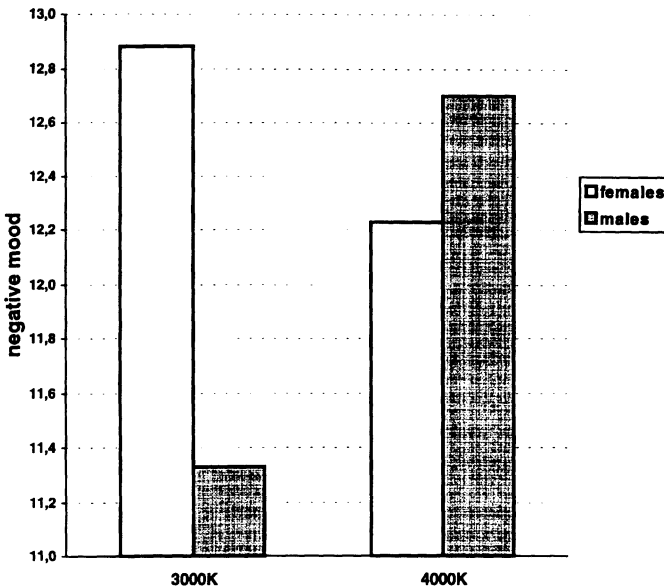


Figure 2: Mean Negative Mood in Females and Males as a Function of 3000K (more reddish) and 4000K (more bluish) Lighting

Problem solving. Subjects' performance scores were subjected to an analysis of variance involving the basic design independent variables. Only the main effect of gender was significant, $F(1, 76) = 5.24, p < .05$. Compared to females, males performed significantly better (21.5 vs. 17.0 out of 32.0). This is consistent with some previous results showing males to be better than females in this type of abstract cognitive task (e.g., Ussher, 1992).

JUDGMENT TASK

For each dimension, subjects' ratings were subjected to a mixed model analysis of variance with the following variables: 2 (color temperatures) $\times$ 2 (female vs. male rater) $\times$ 2 (female vs. male ratee), with the last independent variable as a within-subject factor.

No significant results were obtained involving the color temperature. The following performance appraisal task dimensions showed significant results:

TABLE 1
Mean Appraisal and Cell Standard Deviations (in parentheses) on
Six Dimensions (significant results) as a Function of a Ratee's Gender

<i>Appraisal Dimensions</i>	<i>Female Ratees</i>	<i>Male Ratees</i>	<i>p</i>
Motivation	4.05 (0.94)	3.69 (0.94)	.003
Deserves credit	4.20 (1.08)	3.96 (1.00)	.036
Concern with work-related instructions	4.54 (1.17)	4.15 (1.03)	.005
Work-related achievement	4.41 (0.87)	4.16 (0.70)	.013
Intelligence	4.19 (0.73)	4.35 (0.80)	.046
Overall evaluation	4.30 (0.85)	4.49 (0.75)	.046

Motivation. There was a main effect of ratee, $F(1, 76) = 9.20, p < .01$. As can be seen in Table 1, the female ratee was rated as significantly more motivated than the male, independently of the rater's gender.

Deserves credit. The results showed a main effect of ratee, $F(1, 76) = 4.58, p < 0.05$. Also on this dimension, compared to the male ratee, the female was judged higher (see Table 1). Hence, according to the raters she deserved credit to a higher degree than her male opponent.

Concern with work-related instructions. A main effect of ratee, $F(1, 76) = 8.21, p < 0.01$ again indicated that independently of the rater's gender, the female was rated significantly higher than the male ratee (see Table 1).

Work-related achievement. As on the previous dimension, compared to the male ratee, the female was judged significantly higher, $F(1, 76) = 6.47, p < 0.01$ (see Table 1). Thus, according to the raters, she had a higher level of work-related achievement than her male opponent.

Intelligence. By contrast, on this dimension, the male ratee was rated higher than the female, $F(1, 76) = 4.10, p < .05$ (see Table 1). In addition, the results showed a main effect of rater, $F(1, 76) = 9.23, p < .01$; that is, compared to the male raters, the females judged the ratees as more intelligent, independent of their gender.

Overall evaluation. The results showed one significant result, that of a main effect of ratee, $F(1, 76) = 4.13, p < .05$. The male ratee was evaluated in general as significantly better in terms of performance than the female (see Table 1).

ROOM LIGHT EVALUATION

The subjects' scores on each rating dimension were subjected to an analysis of variance involving the basic design independent variables. The following significant results were obtained:

Glaring. There was a main effect of gender, $F(1, 76) = 7.38, p < .01$, which showed that females perceived the room light as significantly more glaring than did males (see Table 2), independent of the color temperature.

Dim. The results showed one significant result, that of a main effect of gender $F(1, 76) = 11.23, p < .01$. Compared to males, females estimated the room light as significantly less dim (see Table 2).

Soft. There was again a main effect of gender, $F(1, 76) = 8.11, p < .01$. Regardless of the color temperature, females assessed the room light as less soft than did males (see Table 2).

Intense. There was a main effect of gender also on this dimension, $F(1, 76) = 6.15, p < .01$. Females evaluated the room light to be significantly more intense than did males (see Table 2).

Warm. The results showed an interaction between gender and color temperature, $F(1, 76) = 5.20, p < .05$. As can be seen in Figure 3, females evaluated the 3000K lighting as less warm than the 4000K lighting. The opposite result, however, was yielded for males.

Cool. A two-way interaction between gender and color temperature, $F(1, 76) = 5.07$, showed that females estimated the 4000K lighting as less cool than the 3000K lighting, and by contrast, males evaluated the 4000K lighting cooler than the 3000K lighting (see Figure 4).

DISCUSSION

No effect of the lighting on cognitive performance was obtained, as was hypothesized by Knez (1995a). The results supported, however, one important assumption from Knez (1995a), namely, that the artificial light can alter mood. The 4000K lighting preserved best the positive mood and induced the least negative mood in females, and the 3000K lighting accounted for the same result in males (see Figures 1 and 2). Alluding to Boyce (1981, pp. 222-

TABLE 2
Mean Perceived Estimation of the Room Light and Cell Standard
Deviations (in parentheses) on Seven Dimensions as a Function of Gender

<i>Estimation Dimensions</i>	<i>Female Rates</i>	<i>Male Rates</i>	<i>p</i>
Glaring	2.45 (1.25)	1.75 (1.04)	.008
Dim	1.33 (0.69)	1.95 (0.95)	.001
Soft	1.05 (0.22)	1.43 (0.76)	.006
Bright	not significant result		
Warm	not significant result		
Intense	3.53 (1.01)	2.95 (1.05)	.015
Cool	not significant result		

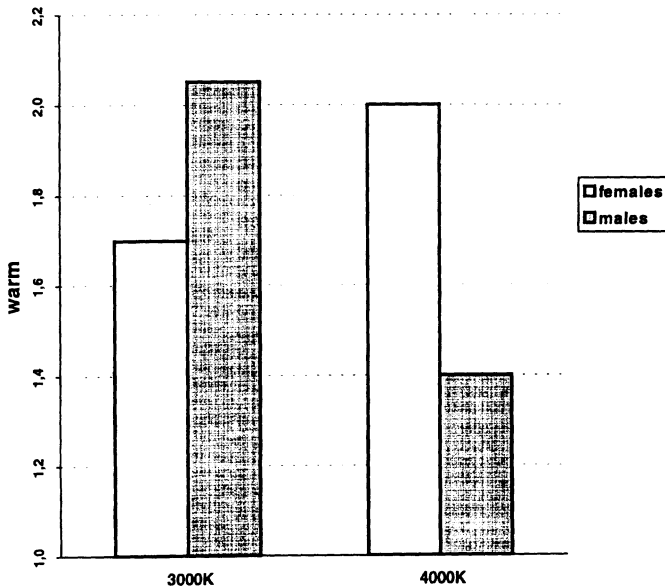


Figure 3: Mean Perceived Estimation of the Room Light in Females and Males on a Warm Dimension and as a Function of 3000K (more reddish) and 4000K (more bluish) Lighting

223), this constitutes experimental evidence for the indirect and nonvisual effects of lighting. Thus, the artificial light is a mood inducer, and the 3000K

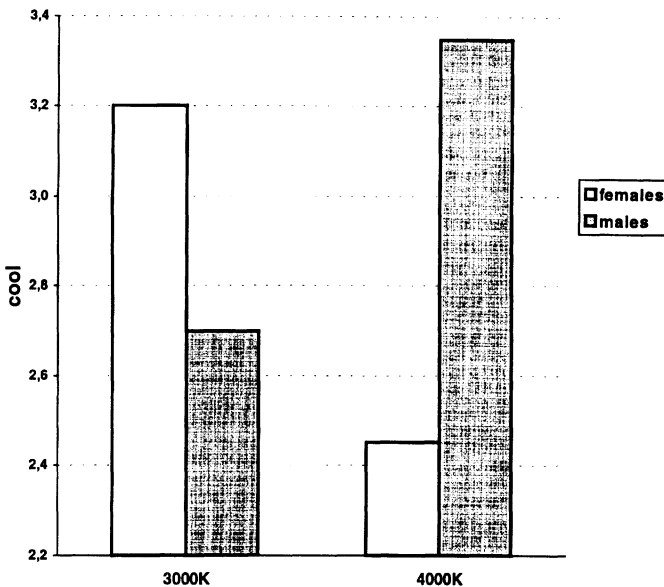


Figure 4: Mean Perceived Estimation of the Room Light in Females and Males on a Cool Dimension and as a Function of 3000K (more reddish) and 4000K (more bluish) Lighting

(more reddish) and 4000K (more bluish) lightings are shown to communicate (Buck, 1984) different affective loadings to and have different affective valances for each gender. This, in turn, suggests speculatively that the coloring quality of an indoor recommended lighting (CIE, 1986) may act as a "mnemonic device" (Rapoport, 1982), triggering culturally specified (in this case gender roles) and affectively based meanings of a color (in general terms), altering mood in genders.

In addition and independent of the lighting exposure, positive mood diminished and negative mood was increased over mood-blocks. This indicates tentatively that the cognitive workload, induced by 2 hours of intellectual work, significantly influenced the subjects' moods.

On four dimensions and consistent with Knez (1995a), the room light was perceived significantly differently by genders. It was considered more glaring, more intense, less dim, and less soft by females than it was by males (see Table 2). Consistent with some previous results in emotional psychology

(Brody & Hall, 1993; Dimberg & Lundquist, 1990; Hall, 1984), these data indicate that females are more expressive (in general terms) toward the indoor lighting (an affective source) than are males.

Moreover, females perceived (between-subject discrimination) the 4000K lighting as most warm, least cool and the 3000K lighting as most cool, least warm. For males, the result yielded an opposite effect, that is, they estimated the 4000K lighting as most cool, least warm and the 3000K lighting as most warm, least cool (see Figures 3 and 4). This suggests that the room light evaluation dimensions of warm and cool are not orthogonal and that the genders are consistent in their discriminations. If we compare this result with the results obtained in the mood section, an interesting congruency appears, namely, the color temperature that preserved best the positive mood and induced the least negative mood in each gender was estimated as significantly most warm and least cool by each gender, respectively. The 4000K lighting accounted for this effect in females, and by contrast, the 3000K lighting yielded this effect for males (compare Figures 1 and 2 with Figures 3 and 4).

Taken together and in line with Knez (1995a, 1995b, 1997) and Knez & Kers (1997), the results question (a) the conclusions of previous results that subjects prefer, in general, more reddish and dim versus more bluish and bright lighting (e.g., Flynn, 1977; Kruithof, 1941) and (b) the present recommendations for the general ceiling-mounted office lighting that do not take individual requirements into consideration (CIE, 1986).

Why are such data present here but not in earlier research? The answer is threefold. First, the extraneous indoor variables such as noise, humidity, and air temperature were carefully controlled, and the lighting system was installed with a high-frequency control of the luminous flux. Second, the lighting exposure lasted for approximately 2 hours. Third, the subjects' moods were measured before and after the exposure.

Finally, and as hypothesized, the results revealed a gender effect in the performance appraisal task that was, except for one result, strictly related to the gender of the ratee—a within-subject effect. In other words, independent of the rater's gender, the female ratee was judged higher on four dimensions than her male opponent, and the reverse result was obtained on two dimensions (see Table 1). Interestingly, despite the female bank clerk's lead in work-related skills, her male opponent was still estimated as more intelligent and was generally (overall evaluation) rated higher. Thus, the raters' processes of judgment were shown to be significantly biased by this demographic variable. Taken together, the consistency of the performance appraisal measure is questioned because it is influenced by the rater's cognitive style (Cardy & Kehoe, 1984), their liking for the ratee (Cardy & Dobbins, 1986), and the

ratee's race (Bigonnes, 1976) and gender (the present study). Therefore, it should be used with caution—particularly in an applied real world context.

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